STAT C205A: Probability Theory

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Fall 2026

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10/29/26 Lecture 17: Hewitt-Savage, Borel-Cantelli, and Convergence in Probability

Theorem 17.1: Hewitt-Savage 0-1 Law

If ξ is an infinite sequence of i.i.d. random variables in $(S, \mathcal{S})$, and $\mathcal{I}$ is the permutation invariant σ-field in $\mathcal{S}^\infty$, then the σ-field $\xi^{-1}(\mathcal{I})$ is P-trivial.

Recall

$$A \triangle B = (A \setminus B) \cup (B \setminus A)$$

and

$$P(A \triangle B) = P(A^c \triangle B^c) = \mathbb{E}\,|1_A - 1_B|\,, \quad A, B \in \mathcal{A}$$

We prove a quick lemma.

Lemma 17.2: Approximation

Given σ-fields $\mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \cdots$ and a set $A \in \bigvee_n \mathcal{F}_n$, there exist $A_1, A_2, \ldots \in \bigcup_n \mathcal{F}_n$ with $P(A \triangle A_n) \to 0$.

Proof: Let $\mathcal{C} = \bigcup_n \mathcal{F}_n$ and let $\mathcal{D}$ denote the class of sets $A \in \bigvee_n \mathcal{F}_n$ with the property. $\mathcal{C}$ is a π-system and $\mathcal{D}$ is a λ-system, so $\bigvee_n \mathcal{F}_n \subseteq \mathcal{D}$.

Proving that $\mathcal{C}$ is a π-system is easy, but proving $\mathcal{D}$ is a λ-system can be hard. For complements, $A \subseteq B$ with $P(A \triangle A_n) \to 0$ and $P(B \triangle B_n) \to 0$,

$$P((B \setminus A) \triangle (B_n \setminus A_n)) \leq P(B \triangle B_n) + P(A \triangle A_n)$$

so we get the desired result.

Now, suppose that $A^j \in \mathcal{D}$ with $A^j \uparrow A$. Let $\epsilon > 0$. Since $A \triangle A^j \downarrow \emptyset$, we see there exists a j such that $P(A \triangle A^j) < \epsilon/2$. Also, since $A^j \in \mathcal{D}$, we can find a $C \in \mathcal{C}$ with $P(A^j \triangle A) < \epsilon/2$, so

$$P(A \triangle C) \leq P(A \triangle A^j) + P(A^j \triangle C) < \epsilon$$

Thus, taking $\epsilon = 1/k$ for each k we can find sets $C^k \in \mathcal{C}$ with $P(A \triangle C^k) \to 0$. Thus, $A \in \mathcal{D}$.

Thus, $\mathcal{D}$ is a λ-system.

■

Now, we can prove the Hewitt-Savage 0-1 Law.

Proof of Hewitt-Savage: Let $\mu = \mathcal{L}(\xi)$. Let $\mathcal{F}_n = \mathcal{S}^n \times S^\infty$ and note $\mathcal{I} \subseteq \mathcal{S}^\infty = \bigvee_n \mathcal{F}_n$. For any $I \in \mathcal{I}$, there exist $B_n \in \mathcal{S}^n$ so that $I_n = B_n \times S^\infty$ satisfies $\mu(I \triangle I_n) \to 0$.

Now, let $\tilde{I}_n = S^n \times B_n \times S^\infty$. By symmetry of μ and I, $\mu\tilde{I}_n = \mu I_n \to \mu I$. Then $\mu(I \triangle \tilde{I}_n) = \mu(I \triangle I_n) \to 0$. Then,

$$\mu(I \triangle (I_n \cap \tilde{I}_n)) \leq \mu(I \triangle I_n) + \mu(I \triangle \tilde{I}_n) \to 0$$

Since $I_n \perp\!\!\!\perp \tilde{I}_n$ under μ,

$$\mu I \leftarrow \mu(I_n \cap \tilde{I}_n) = (\mu I_n)(\mu \tilde{I}_n) \to (\mu I)^2$$

so $\mu I = 0$ or 1, so $P(\xi^{-1}(I)) = 0$ or 1.

■

Corollary 17.3: Random Walk

Let $\xi_1, \xi_2, \ldots$ be i.i.d. non-degenerate random variables, and let $S_n = \xi_1 + \cdots + \xi_n$. Then,

(i) $P\{S_n \in B \text{ i.o.}\} = 0$ or 1 for any $B \in \mathcal{B}$.

(ii) $\limsup_n S_n = \infty$ a.s. or $-\infty$ a.s.

(iii) $\limsup_n (\pm S_n) = \infty$ if the ξ_n are symmetric; that is, $\xi_n \overset{d}{=} -\xi_n$.

Proof: **(i)** $\{S_n \in B \text{ i.o.}\}$ is permutation invariant in the sense for HS.

(ii) We know $\limsup_n S_n = c$ a.s. by HS and a.s. degeneracy. Then, as

$$c = \limsup_n S_{n+1} = \limsup_n (S_{n+1} - \xi_1) + \xi_1 = c + \xi_1$$

so if $|c| < \infty$, then $\xi_1 = 0$ a.s., which contradicts non-degeneracy of ξ_1. Thus, $|c| = \infty$.

(iii) We see

$$c = \limsup_n S_n \geq \liminf_n S_n = -\limsup_n (-S_n) = -c$$

so $-c \leq c \in \{\pm\infty\}$, so $c = \infty$.

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Theorem 17.4: Borel-Cantelli

Let $A_1, A_2, \ldots \in \mathcal{A}$. Then

(i) $\sum_n PA_n < \infty$ implies $P\{A_n \text{ i.o.}\} = 0$.

(ii) If the A_n are independent, then $\sum_n PA_n = \infty \implies P\{A_n \text{ i.o.}\} = 1$

Proof: **(i)**

$$\mathbb{E}\left(\sum_n 1_{A_n}\right) = \sum_n \mathbb{E}1_{A_n} = \sum_n PA_n < \infty$$

so $\sum_n 1_{A_n} < \infty$ almost surely. Thus, $P\{A_n \text{ i.o.}\} = 0$.

(ii) Assume the A_n are independent and $\sum_n PA_n = \infty$. Using that $1 - x \le e^{-x}$ for all x,

$$\begin{aligned} P\left(\bigcup_{k\ge n} A_k\right) &= 1 - P\left(\bigcap_{k\ge n} A_k^c\right) \\ &= 1 - \prod_{k\ge n} PA_k^c \\ &= 1 - \prod_{k\ge n} (1 - PA_k) \\ &\ge 1 - \prod_{k\ge n} e^{-PA_k} \\ &= 1 - e^{-\sum_{k\ge n} PA_k} = 1 - e^{-\infty} = 1 \end{aligned}$$

Hence, as $n \to \infty$,

$$P\{A_n \text{ i.o.}\} = \lim_{n\to\infty} P\left(\bigcup_{k\ge n} A_k\right) = 1$$

■

Example: Independence is Needed

Let $U \sim \text{Unif}(0,1)$ and $A_n = \{U \in (0, 1/n)\}$. Note we see $P(A_n) = 1/n$, so $\sum_n PA_n = \infty$, but $P\{A_n \text{ i.o.}\} = 0$. Thus, independence is important.

Lemma 17.5: Bienaymé-Chebyshev-Paley-Zygmund Inequality

Let ξ be an $\mathbb{R}_+$-valued random variable with $0 < \mathbb{E}\xi < \infty$. Then,

$$\frac{(\mathbb{E}\xi)^2}{\mathbb{E}[\xi^2]} \left[(1-r)_+\right]^2 \le P\{\xi > r\mathbb{E}\xi\} \le \frac{1}{r}, \quad r \ge 0$$

The upper bound is also known as Markov's inequality.

Proof: Assume WLOG $\mathbb{E}\xi = 1$ (we can redefine $\eta = \xi/\mathbb{E}\xi$).

Since $r1\{\xi \ge r\} \le \xi$, we see that $rP\{\xi \ge r\} \le \mathbb{E}[\xi]$, so $P\{\xi \ge r\} \le \frac{\mathbb{E}[\xi]}{r}$, and just redefine r.

To get the lower bound, note

$$\begin{aligned} 1 = \mathbb{E}[\xi] &= \mathbb{E}[\xi 1\{\xi \le r\}] + \mathbb{E}[\xi 1\{\xi > r\}] \\ &\le r + \mathbb{E}[\xi^2]^{1/2} P\{\xi > r\}^{1/2} \end{aligned}$$

so for $r' := r\mathbb{E}[\xi]$,

$$\begin{aligned}\mathbb{E}[\xi] &\le r\mathbb{E}[\xi] + \mathbb{E}[\xi^2]^{1/2} P\{\xi > r\mathbb{E}[\xi]\}^{1/2} \\ \mathbb{E}[\xi](1-r) &\le \mathbb{E}[\xi^2]^{1/2} P\{\xi > r\mathbb{E}\xi\}^{1/2} \\ \frac{(\mathbb{E}[\xi])^2}{\mathbb{E}[\xi^2]} [(1-r)_+]^2 &\le P\{\xi > r\mathbb{E}\xi\}\end{aligned}$$

■

Definition 17.6: Convergence in Probability

Given random elements ξ and $\xi_1, \xi_2, \dots$ in a metric space (S, ρ), say that ξ_n *converges in probability* to ξ, written $\xi_n \xrightarrow{P} \xi$, if

$$\lim_{n\to\infty} P\{\rho(\xi_n, \xi) > \epsilon\} = 0, \quad \epsilon > 0$$

By Chebyshev's inequality, this is equivalent to $\mathbb{E}[\rho(\xi_n, \xi) \wedge 1] \to 0$. This is because, for $\epsilon < 1$

$$P\{\rho(\xi_n, \xi) > \epsilon\} = P\{\rho(\xi_n, \xi) \wedge 1 > \epsilon\} \le \frac{\mathbb{E}[\rho(\xi_n, \xi) \wedge 1]}{\epsilon}$$

aso the LHS goes to zero as $\mathbb{E}[\rho(\xi_n, \xi) \wedge 1] \to 0$. Similarly, because

$$\mathbb{E}[\rho(\xi_n, \xi) \wedge 1] \le \epsilon + P\{\rho(\xi_n, \xi) > \epsilon\}$$

so taking $n \to \infty$, we see that $\lim_{n\to\infty} \mathbb{E}[\rho(\xi_n, \xi) \wedge 1] \le \epsilon$ for all ϵ.

Here is a more useful definition.

Lemma 17.7: Subsequence Criterion

Let $\xi, \xi_1, \xi_2, \dots$ be random elements in a metric space (S, ρ). Then $\xi_n \xrightarrow{P} \xi$ iff every subsequence ξ_{n_m} has a further subsequence $\xi_{n_{m_k}}$ such that $\xi_{n_{m_k}} \to \xi$ a.s. In particular, $\xi_n \to \xi$ a.s. implies $\xi_n \xrightarrow{P} \xi$.

Proof: ($\Rightarrow$) Assume $\xi_n \xrightarrow{P} \xi$ and let ξ_{n_m} be a subsequence. Since $\mathbb{E}[\rho(\xi_{n_m}, \xi) \wedge 1] \to 0$, pick a further subsequence so that $\mathbb{E}[\rho(\xi_{n_{m_k}}, \xi) \wedge 1] < 2^{-k}$. Then

$$\mathbb{E} \sum_k \{\rho(\xi_{n_{m_k}}, \xi) \wedge 1\} \overset{\text{MCT}}{=} \sum_k \mathbb{E}[\rho(\xi_{n_{m_k}}, \xi) \wedge 1] < \infty$$

Then, $\sum_k \rho(\xi_{n_{m_k}}, \xi) \wedge 1$ is finite a.s., so $\rho(\xi_{n_{m_k}}, \xi) \to 0$ a.s.

($\Leftarrow$) Assume the subsequence condition. Now, if $\xi_n \not\xrightarrow{P} \xi$, there exists $\epsilon > 0$ such that $\mathbb{E}[\rho(\xi_{n_m}, \xi) \wedge 1] > \epsilon$ along a subsequence ξ_{n_m}. By hypothesis, we know $\xi_{n_{m_k}} \to \xi$ a.s. for some further subsequence, so by DCT, we know

$$\mathbb{E}[\rho(\xi_{n_{m_k}}, \xi) \wedge 1] \to 0$$

which is a contradiction.

■

Lemma 17.8: Continuous Mapping

For any metric spaces S, T, let $\xi, \xi_1, \xi_2, \ldots$ be random elements in S with $\xi_n \xrightarrow{P} \xi$ and let $f : S \to T$ be measurable and a.s. continuous at ξ. Then, $f(\xi_n) \xrightarrow{P} f(\xi)$.

Proof: Let ξ_{n_m} be a subsequence. Then, find a further subsequence $\xi_{n_{m_k}} \to \xi$ a.s. Then, by continuity,

$$f(\xi_{n_{m_k}}) \to f(\xi) \quad \text{a.s.}$$

Thus, $f(\xi_{n_m})$ has an a.s. convergent subsequence, so $f(\xi_n) \xrightarrow{P} f(\xi)$ by above. ■

Consider a sequence of metric spaces (S_k, ρ_k) and consider the product space $S = \bigtimes_k S_k = S_1 \times S_2 \times \cdots$ with the product topology. It turns out a convenient metrization is given by

$$\rho(x, y) = \sum_k 2^{-k} \{\rho_k(x_k, y_k) \wedge 1\}, \quad x, y \in \bigtimes_k S_k$$

Note for $x, x^{(1)}, x^{(2)}, \ldots \in \bigtimes_k S_k$, $\lim_n \rho(x^{(n)}, x) = 0$ iff $\lim_n \rho_k(x_k^{(n)}, x_k) = 0$ for all k.

If each S_k is separable, recall $\mathcal{B}(S) = \bigotimes_k \mathcal{B}(S_k)$, so a random element in S is simply a sequence of random elements in S_k, $k \in \mathbb{N}$.

Lemma 17.9: Random Sequences

For any separable metric spaces $S_1, S_2, \ldots$, let $\xi = (\xi_1, \xi_2, \ldots)$ and $\xi^n = (\xi_1^n, \xi_2^n, \ldots)$, $n \in \mathbb{N}$ be random elements in $\bigtimes_k S_k$. Then,

$$\xi^n \xrightarrow{P} \xi \quad \text{iff} \quad \xi_k^n \xrightarrow{P} \xi_k \text{ in } S_k \text{ for each } k$$

Corollary 17.10: Elementary Operations

Let $\xi, \xi_1, \xi_2, \ldots$ and $\eta, \eta_1, \eta_2, \ldots$ be random variables with $\xi_n \xrightarrow{P} \xi$ and $\eta_n \xrightarrow{P} \eta$. Then,

(i) $a\xi_n + b\eta_n \xrightarrow{P} a\xi + b\eta$ for all $a, b \in \mathbb{R}$.

(ii) $\xi_n \eta_n \xrightarrow{P} \xi\eta$.

(iii) $\xi_n / \eta_n \xrightarrow{P} \xi/\eta$ whenever $\eta \neq 0$ and $\eta_n \neq 0$ a.s. for all n.

Proof: $(\xi_n, \eta_n) \xrightarrow{P} (\xi, \eta)$, and as the operations are continuous, it follows. ■